



SES-12 satellite launched

A used SpaceX Falcon 9 rocket launched an SES-12 communications satellite into orbit
<https://digit.in/jul18-77-1>



NASA reinvents the wheel

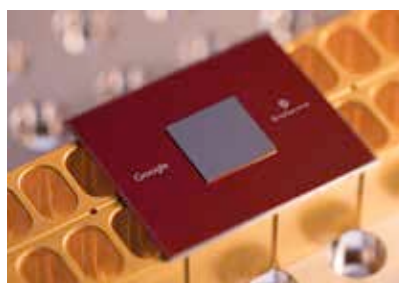
NASA is reinventing the wheel for space by making shape memory alloy tires to adapt to terrains
<https://digit.in/jul18-77-2>

put the time period into perspective, the fastest thing in the universe, light, takes more than a femtosecond to cross a single red blood cell. So, how does this phenomenon even matter for computing? It does when successive beams of light can knock them back and forth between the two excited tracks, effectively creating a string of 1s and 0s in spectacularly quick flashes.

HOW IT WORKS

We've reached a point where quantum computers in laboratories around the world are working with 50 or more qubits. Earlier this year, in March 2018, Google announced a quantum computer with 72 qubits! That is well beyond what the most powerful traditional computer can simulate, which could be used to insinuate that humans have already achieved quantum supremacy. Yet, we have not seen any celebratory announcements or declarations from the quantum computing community yet. Why would that be?

That is because quantum computing's basic units, the qubits, are notoriously difficult to maintain. As we mentioned earlier, electrons aren't exactly keen on maintaining a particular state for too long. Even with Google's latest quantum computing platform Bristlecone, quantum computers need to be kept extremely cold and physically stable to avoid errors. To put a parameter on it, a quantum computer needs to have an error rate of less than 0.5 percent for every two qubits to be useful in even remotely



Google's Bristlecone can handle 72 qubits

real world scenarios. The best Google has achieved so far is 0.6 percent with their 9 qubit linear array. And we're speculating about 72 qubits here which is definitely an insurmountable barrier in the near future.

The results achieved with the tungsten-selenium lattice could be the solution. These results have been achieved in room temperature conditions and the researchers have speculated that the same could be put to use for quantum computing. This is because this approach differs from trying to hold on to a quantum state for longer - it essentially tries to increase the amount of processing that can be done during the same amount of time, akin to increasing the clock speed on traditional computers. This can be done due the ultrashort pulses of light that can be supplied.

These extremely short, configurable bursts of light were demonstrated almost a year ago by researchers, including engineers from the University of Michigan. They showed that the peaks within the laser could be controlled and could also be twisted,

hence their ability to affect the quantum state of the electron differently each time they are used to excite it. The world of physicists would call this 'lightwave electronics'.

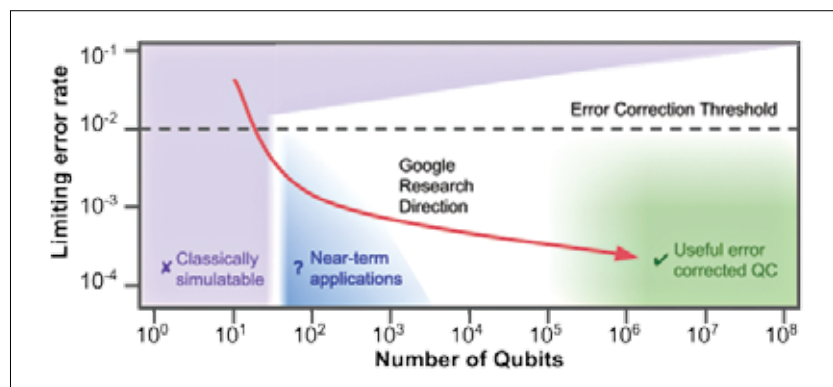
Along with the two distinct states before and after the excitation, the electrons can also occupy superposition states where they hold both values 1 and 0 at the same time. Additionally, once energy is supplied, the electrons are constantly emitting light, making it easier to read the qubit without disturbing the already fragile quantum state.



IBM's 50 qubit system

THE SIGNIFICANCE

Even though the current results hardly venture into actual quantum computing, in theory they do work for the area. The researchers have successfully demonstrated this operation with one qubit and the next logical step would involve adding another qubit and trying to get them to interact meaningfully. "In the long run, we see a realistic chance of introducing quantum information devices that perform operations faster than a single oscillation of a lightwave," said Rupert Huber, professor of physics at the University of Regensburg, who led the experiment. "The material is relatively easy to make, it works in room temperature air, and at just a few atoms thick, it is maximally compact." 



2D conceptual chart showing the relationship between error rate and number of qubits. The intended research direction of Google's Quantum AI Lab is shown in red